

Community Hero

Hyperlocal Problem Solver — An AI-Powered Civic Engagement Platform

VIBE2SHIP · Coding Ninjas × Google for Developers · Powered by Google Gemini
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Executive Summary

Community Hero turns every citizen with a smartphone into a sensor for their city. A resident snaps a photo of a pothole, water leak, broken streetlight or hazard; **Google Gemini** instantly identifies the problem, scores its severity and routes it to the right municipal department; neighbours **verify** it with a tap; authorities act on an **AI-prioritised queue**; and everyone tracks progress to resolution — transparently, in real time.

It is a complete, working, deployed product that directly answers the *Community Hero – Hyperlocal Problem Solver* brief, with **Google Gemini (Google AI Studio)** — the hackathon’s mandated core tool — as its intelligence layer.

Live demo: <https://hitotsume-nozo-hephasteus.hf.space> **Source code:**
<https://github.com/Hitotsume-nozo/community-hero>

The Problem

Communities everywhere face the same civic frustrations — potholes, water leakages, damaged streetlights, garbage, drainage failures and public-safety hazards. Yet reporting these issues today is:

- **Fragmented** — scattered across phone helplines, WhatsApp groups and paper complaints.
- **Opaque** — citizens rarely learn whether anyone received, prioritised or acted on a report.
- **Untracked** — there is no shared, verifiable record from report to resolution.
- **Low-trust** — without visible accountability, participation withers.

The result: problems linger, the same complaint is filed many times, and authorities lack the clean, prioritised data they need to allocate scarce crews efficiently.

Our Solution

Community Hero is a single, mobile-first platform that closes the entire civic loop — **identify** → **report** → **validate** → **track** → **resolve** — and makes every step transparent and collaborative. Artificial intelligence removes the friction at each stage:

- **Reporting** takes seconds — point, shoot, submit. Gemini writes the title, description and category for you.
- **Validation** is crowdsourced — three neighbour confirmations auto-verify a report, filtering noise and surfacing what truly matters.
- **Tracking** is live — a real-time feed and a full status timeline keep citizens informed.
- **Resolution** is data-driven — authorities work a queue ranked by severity, community weight and SLA urgency, with Gemini forecasting where the next problems will emerge.

Key Features (mapped to the evaluation criteria)

Evaluation focus	How Community Hero delivers
Image / video reporting	One-tap photo capture; Gemini Vision auto-detects the issue and pre-fills the entire form.
AI-powered categorization	12 civic categories, severity 1–5, safety-risk rating, department routing and an estimated resolution SLA — each with a confidence score.
Geo-location & mapping	Interactive clustered map of the city, a severity heatmap , GPS capture and a draggable location pin.
Community verification	“I see this too” confirmations; a report auto-verifies once three neighbours vouch for it.
Real-time issue tracking	Live updates via Server-Sent Events

Evaluation focus	How Community Hero delivers
	and a complete status lifecycle timeline (Reported → Verified → In Progress → Resolved).
Impact dashboards	KPIs, a 30-day trend chart, status breakdown, and category & ward leaderboards.
Predictive insights	Gemini forecasts emerging hotspots and gives authorities prioritised, actionable resource recommendations.
Gamification	Civic points, levels, badges and a city-wide “Community Heroes” leaderboard.

Going Beyond the Brief

- **Geo + category duplicate detection** — before a report is filed, the app checks for similar nearby issues and nudges the citizen to confirm the existing one, keeping data clean.
- **Gemini Civic Assistant** — a floating chat assistant that answers natural-language questions (“Which ward needs the most attention?”) grounded on **live** city data.
- **AI complaint drafting** — one tap generates a formal, department-addressed complaint letter the citizen can copy into an official grievance portal.
- **Authority Console** — a dedicated triage workspace with an AI-prioritised queue, SLA / overdue tracking and one-click status updates.
- **Resilience by design** — every AI capability has a deterministic fallback, so the platform stays fully usable even if the AI service is unavailable.

How Google Gemini Powers the Platform

All AI runs server-side via the Google AI Studio Gemini API (model `gemini-2.5-flash`, multimodal), using enforced structured (JSON-schema) output for reliability.

Capability	What Gemini does
Multimodal triage	Analyses the uploaded photo (and any note) and returns category, title, description, severity, safety-risk, tags and a confidence score — including an “is this actually a civic issue?” guard.
Predictive insights	Reads aggregated ward × category data and forecasts the hotspots most likely to worsen, with prioritised recommendations for authorities.
Civic assistant	Answers citizen questions using live issue statistics as grounding context.
Complaint drafting	Writes a polite, firm, time-bound complaint letter addressed to the correct department.

To keep the live experience fast and dependable, “thinking” tokens are disabled for these structured calls, and any API hiccup transparently falls back to a heuristic — clearly labelled in the interface so trust is never compromised.

Architecture

Community Hero ships as a **single deployable container** for simplicity and reliability.

Layer	Technology	Responsibility
Client	React + TypeScript + Vite	Map, AI report flow, dashboards, console and assistant UI
Server	Node + Express + TypeScript	REST API, real-time event stream, gamification, duplicate detection
Intelligence	Google Gemini (AI Studio)	Multimodal triage, prediction, assistant, drafting
Realtime	Server-Sent Events	Live issue and status

Layer	Technology	Responsibility
		updates across all clients
Persistence	JSON document store	Zero-provisioning storage of issues and users
Mapping	Leaflet + OpenStreetMap	Keyless, always-available maps, clustering and heat

The Express server also serves the built React application, so the entire product runs from one origin with no external infrastructure to provision — making it inexpensive and trivial to host.

Technology Stack

- **Frontend:** React 18, TypeScript, Vite, Tailwind CSS, React-Leaflet (clustering + heatmap), Recharts, Framer-Motion.
- **Backend:** Node 22, Express, TypeScript, Server-Sent Events.
- **AI:** Google Gemini via Google AI Studio (multimodal, structured output).
- **Maps:** Leaflet with OpenStreetMap / CARTO tiles (no API key required).
- **Deployment:** Docker container on Hugging Face Spaces (free); also one-command deployable to Google Cloud Run or Render.

Impact, Transparency & Engagement

Community Hero is designed to build a virtuous civic cycle:

- **Transparency** — every report has a public status timeline; the impact dashboard exposes resolution rates and average fix times for the whole city.
- **Accountability** — issues are routed to named departments with visible SLAs; overdue items are flagged automatically.
- **Participation** — gamification (points, levels, badges, leaderboard) rewards citizens for reporting, verifying and following through, sustaining engagement over time.
- **Efficiency** — AI prioritisation and hotspot prediction help authorities deploy limited crews where they matter most.

Live Demo Walkthrough

1. **Sign in** and land on a live map of the city seeded with realistic issues.
2. **Report** — upload a photo; watch Gemini auto-fill category, severity and department; place the pin (duplicate detection runs); submit and earn civic points.
3. **Issue detail** — confirm an issue, see it auto-verify, and generate an AI complaint letter.
4. **Impact dashboard** — explore KPIs, trends and Gemini's predicted hotspots.
5. **Authority console** — switch roles and work the AI-prioritised queue from In Progress to Resolved.
6. **Civic Assistant** — ask a question and get a data-grounded answer.

Running & Deploying

The project runs locally with a single command set (`npm install` then `npm run build && npm start`) and deploys to a free public link on Hugging Face Spaces. Detailed, copy-paste instructions for Hugging Face, Render and Google Cloud Run are included in the repository's `START_HERE.md` and `DEPLOY.md`.

Future Scope

- Native mobile apps with offline-first capture and background sync.
- Direct integration with municipal grievance systems and open-data portals.
- Multilingual reporting and assistant responses for wider inclusion.
- Verified authority workflows with audit logs and citizen satisfaction feedback at closure.
- Video and audio evidence with Gemini multimodal understanding.

Submission Details

- **Track / Problem Statement:** Community Hero – Hyperlocal Problem Solver.
- **Event:** VIBE2SHIP — Coding Ninjas × Google for Developers.
- **Core tool:** Google AI Studio (Gemini) — central to triage, prediction, assistance and drafting.
- **Live application:** <https://hitotsume-nozo-hephasteus.hf.space>

- **Repository:** <https://github.com/Hitotsume-nozo/community-hero>
- **Participation:** Solo.